

DisasterPrepEdu: A Scalable Web-Based Disaster Preparedness Training Platform for Educational Institutions

Lisha S
Dept. of Computer Science
Engineering Presidency
University
Rajankunte, India
lisha20221cse0534@gmail.com

Nidhi Bhushan H
Dept. of Computer Science
Engineering Presidency
University
Rajankunte, India
nidhibhushanh@gmail.com

Kalpana k Harish
Assistant Professor
Dept. of Computer Science
Engineering Presidency
University
Rajankunte, India
kalpanaharish1983@gmail.com

Rohith SL
Dept. of Computer Science
Engineering Presidency
University Rajankunte, India
rohithlohith0310@gmail.com

ABSTRACT

India experiences recurrent natural disasters including earthquakes, floods, and fires, yet preparedness education within schools and colleges is largely confined to theoretical guidelines, with limited emphasis on practical training. This paper introduces **DisasterPrepEdu**, a web-based disaster preparedness education platform designed to bridge this gap by enabling students to acquire actionable response skills. The proposed system adopts a structured three-phase learning approach consisting of instructional content delivery, knowledge evaluation, and time-bound virtual disaster simulations. Gamification elements are incorporated to improve learner engagement, while role-based access ensures a clear separation between student training activities and administrative monitoring of preparedness and compliance. The platform supports adaptable, region-aware disaster curricula and extends its applicability across both K-12 and higher education institutions, overcoming the narrow scope of existing child-centric awareness tools. By replacing infrequent physical drills with scalable digital training, the system reduces logistical challenges for institutions. Preliminary internal validation suggests improved awareness of disaster response procedures and indicates the system's suitability for broader academic deployment, supporting the practical implementation of national disaster risk reduction initiatives.

Keywords- Disaster preparedness education, gamified learning, virtual simulations, school safety, role-based access, disaster risk reduction, educational technology

I. INTRODUCTION

India is among the world's most disaster-vulnerable nations, experiencing frequent earthquakes, seasonal flooding, and recurring fire incidents that pose significant risks to educational institutions. Large regions of the country fall within seismic Zones II-IV, while annual monsoon flooding affects millions of individuals, including school and college

populations. Despite the availability of comprehensive safety and preparedness guidelines issued by the National Disaster Management Authority (NDMA), disaster preparedness education within academic institutions remains largely theoretical, with limited emphasis on sustained practical training.

Existing preparedness efforts in schools and colleges are typically confined to infrequent manual drills and awareness sessions, which are often difficult to coordinate and inadequately reinforced over time. As a result, students and staff may lack the confidence and procedural knowledge required to respond effectively during emergency situations. This gap between policy-level guidance and practical skill development highlights the need for scalable, technology-driven disaster education frameworks.

This paper presents **DisasterPrepEdu**, a web-based disaster preparedness education platform designed to translate NDMA-aligned guidelines into actionable response skills for K-12 and higher education learners. The proposed system adopts a structured three-phase methodology consisting of instructional content delivery, knowledge assessment, and time-constrained virtual simulations. By replacing sporadic physical drills with continuous digital practice, the platform reduces logistical constraints while maintaining alignment with institutional safety curricula.

The key technical contributions of this work are summarized as follows:

1. A three-phase progressive training architecture enabling continuous disaster preparedness learning.
2. A role-based dual-interface system supporting gamified student learning and administrative preparedness monitoring.
3. A region-adaptive disaster curriculum configurable to location-specific hazard profiles.
4. A scalable web-based infrastructure suitable for deployment across diverse educational institutions without specialized hardware.

The remainder of this paper is organized as follows. Section II reviews related work and limitations of existing digital disaster education platforms. Section III describes the proposed system architecture and workflow. Section IV details the implementation of the platform, and Section V presents internal validation results and discusses system readiness for institutional deployment.

II. RELATED WORK

A. Related Work in Digital Disaster Education

Digital disaster preparedness education has evolved as a complement to traditional drills and classroom lectures. Early platforms focused primarily on children, using quizzes and interactive content to teach basic disaster concepts. Rodavia *et al.* [1] developed a quiz-based platform for elementary students that included videos and simple games. While effective in raising awareness, it lacked administrative tools and coverage for older learners. Similarly, Hassan *et al.* [2] proposed web-based modules for a general audience, offering interactive content but without scenario-based simulations or local hazard customization. These studies highlighted the potential of digital solutions but revealed gaps in interactivity, age coverage, and practical skill-building.

Later research explored augmented reality (AR) and immersive simulations to provide more realistic training. Wang *et al.* [3] implemented an AR evacuation system for college students, enabling them to practice emergency responses in real time. Lu *et al.* [4] developed a collaborative AR platform for elementary students to facilitate group learning. Tsai *et al.* [5] designed an AR-based earthquake simulation, providing scenario-based training experiences. While these systems improved engagement, they often required specialized hardware and focused on a single hazard, which limited widespread adoption.

Gamification approaches have also been applied to increase learner motivation and knowledge retention. Rahman *et al.* [6] and the Indonesian Flood Game [12] demonstrated improved retention and engagement through serious games, while general studies [11] highlighted the benefits of badges and leaderboards. However, most gamified platforms targeted awareness rather than hands-on skill development and were limited to specific hazards or regions. Teacher- and professional-focused e-learning platforms, such as ELITE-DR [7] and the Web 2.0 disaster education program [8], focused on cognitive knowledge acquisition but lacked scenario-based exercises, student-centered design, and administrative oversight. Public applications, including FEMA [14] and NDMA mobile apps [15], offered alerts and basic guidance but did not provide structured curricula, gamification, or assessment suitable for formal educational settings.

A summary of the most relevant platforms is provided in Table I, highlighting their key features and limitations.

Table 1. Comparison of Existing Digital Disaster Education Platforms

Ref	Platform Type	Target Audience	Key Features	Limitations
[1]	Quiz app	Elementary	Videos, basic games	Child-only, no admin tools
[2]	Web modules	General	Interactive content	No simulations, generic content
[3]	AR evacuation	College	Response timing	Single hazard focus
[6]	Gamified flood game	School	Retention, engagement	Region-specific, awareness only
[7]	E-learning DRM	Medical students	Self-learning videos	No practical exercises
[12]	Serious game	School	Retention, interactive	Single disaster type

From Table I, it is evident that existing platforms often have limitations in coverage, practical training, and institutional oversight, which motivates the development of a more comprehensive system.

B. Limitations of Existing Digital Disaster Education Platforms

A careful review of the literature reveals several recurring limitations in existing platforms:

1. **Limited User Coverage:** Most platforms target either young learners or professionals, leaving K–12 and higher education underserved.
2. **Single-Disaster Focus:** Platforms often address only one type of hazard (e.g., floods, earthquakes), reducing applicability across diverse disaster scenarios.
3. **Hardware Dependency:** AR and VR-based systems require specialized equipment, limiting scalability and institutional adoption.
4. **Lack of Administrative Oversight:** Few systems provide dashboards for tracking student progress, compliance, or institutional preparedness.
5. **Insufficient Practical Training:** Most solutions emphasize awareness and cognitive learning rather than hands-on, scenario-based skill development.
6. **Region-Specific Constraints:** Existing platforms often lack adaptation to local hazard profiles, reducing relevance for particular geographic areas.
7. **Fragmented Integration:** Few systems combine progressive learning, gamification, simulations, and administrative tools into a unified platform.

These limitations highlight the need for a comprehensive, web-based, role-based disaster education system. The proposed DisasterPrepEdu platform addresses these gaps by integrating progressive training modules, gamified engagement, virtual drills, region-adaptive curricula, and administrative monitoring, providing scalable disaster preparedness education across K–12 and higher education institutions.

III. SYSTEM ARCHITECTURE AND WORKFLOW

A. System Overview

The proposed *DisasterPrepEdu* platform is designed as a web-based disaster preparedness training system that supports structured learning, assessment, and emergency response simulation within a single unified environment. The architecture follows a layered design to ensure scalability, accessibility, and efficient institutional deployment. The overall system architecture is illustrated in Fig. 1.

The architecture consists of three primary layers: the Presentation Layer, Application Layer, and Data Layer. The **Presentation Layer** provides a web interface through which users interact with the platform. Students access learning modules, quizzes, and virtual drills, while teachers and administrators use analytical dashboards to monitor participation, performance, and preparedness levels. Role-based access control ensures that each user interacts only with authorized functionalities.

The **Application Layer** forms the operational core of the platform. It manages user authentication and role verification, learning module delivery, quiz evaluation, emergency drill execution, analytics processing, and gamification logic. Specialized services within this layer handle real-time feedback, leaderboard updates, and achievement tracking. The application layer may also read optional external disaster-related data to enhance scenario realism and regional relevance.

The **Data Layer** is responsible for persistent storage and management of system data, including user profiles, learning progress, quiz scores, drill results, and achievement records. A centralized database structure ensures data consistency, secure access, and efficient retrieval for large-scale institutional usage.

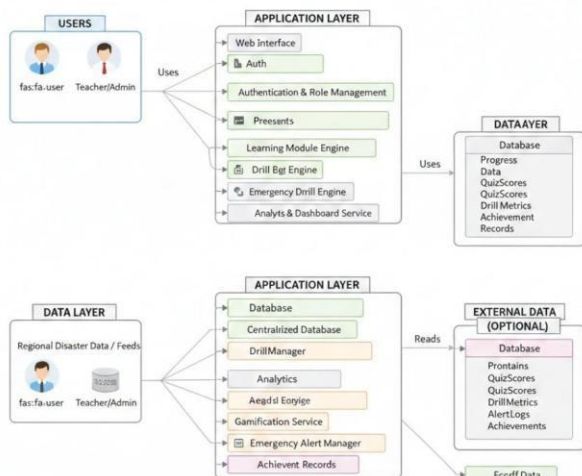


Fig. 1. System Architecture of the DisasterPrepEdu Platform

Fig. 1 illustrates the layered system architecture highlighting user interaction through a web interface, core application services such as authentication, learning modules, emergency drill engines, analytics processing, and gamification services, along with centralized data storage and optional integration of external disaster information sources.

B. User Workflow and Interaction Model

The user interaction and role-based workflow of the proposed system is shown in Fig. 2. The workflow begins with user registration or login, followed by authentication and role identification. Based on the assigned role, the system directs users to either the student workflow or the administrator workflow.

In the **student workflow**, users access a personalized dashboard that presents structured learning modules focused on disaster preparedness concepts. After completing instructional content, students attempt quizzes designed to assess conceptual understanding. Successful quiz completion unlocks access to virtual disaster drills, where users participate in time-bound simulated emergency scenarios. Real-time feedback, performance scores, and leaderboard rankings are provided to enhance engagement and reinforce learning outcomes.

In the **administrator workflow**, users access a dedicated dashboard that provides an overview of student participation, assessment results, and drill engagement. Administrators can monitor real-time analytics, review simulation outcomes, and evaluate preparedness levels across individuals or institutions. This workflow supports institutional oversight, training effectiveness evaluation, and compliance reporting.

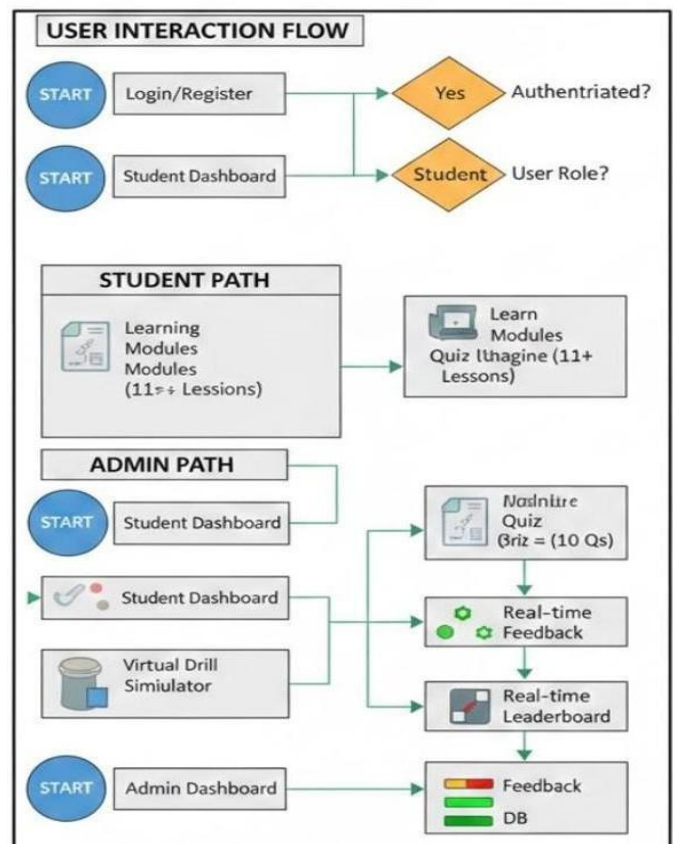


Fig. 2. User Interaction and Role-Based Workflow

Fig. 2 depicts the sequential user interaction flow beginning with authentication and role verification, followed by branching into student and administrator paths. The student path includes learning modules, quizzes, and virtual drill simulations with real-time feedback and leaderboard updates, while the administrator path focuses on monitoring, analytics, and preparedness management.

IV. IMPLEMENTATION

This section describes the practical realization of the proposed *DisasterPrepEdu* platform. The implementation focuses on modularity, role-based access, and scalability to support continuous disaster preparedness training across educational institutions.

A. Platform Implementation and Functional Modules

The *DisasterPrepEdu* system was implemented as a modular, role-based web application to support structured learning, assessment, and simulation-based training. After authentication, users are directed to role-specific dashboards that serve as the primary interaction point for all platform activities.

For students, the dashboard aggregates key preparedness indicators such as completed learning modules, quiz performance, virtual drill participation, earned badges, and cumulative preparedness scores. These indicators are dynamically updated based on user activity, allowing learners to monitor their progress and continue training seamlessly.

The learning module subsystem delivers structured content covering multiple disaster types, including earthquakes, floods, and fire emergencies. Each module consists of instructional lessons followed by quizzes to assess conceptual understanding. Sequential access control ensures that learners complete required lessons and assessments before progressing further, maintaining alignment with the proposed three-phase learning methodology.

Gamification mechanisms are integrated into the application logic to enhance learner engagement and retention. Points, badges, learning streaks, and leaderboard rankings are automatically computed based on quiz results, drill performance, and consistency of participation. These elements are updated in real time without interrupting the learning workflow.

The virtual drill module provides time-bound simulated disaster scenarios, such as earthquake evacuation exercises. During each drill, user actions, response accuracy, and completion time are recorded to generate drill-specific performance metrics. These metrics contribute to both individual preparedness scores and institutional analytics.

An emergency contacts module is included to support immediate response during real-world situations. It provides quick access to essential emergency services such as police, fire, medical, and campus authorities through categorized listings and one-click interaction. This module remains accessible independently of training progress to prioritize safety.

The **progress reporting and analytics subsystem** consolidates learning activity, assessment outcomes, and drill participation into visual analytics such as weekly activity trends, points distribution, and average performance indicators. These analytics support both learners and administrators in identifying preparedness growth and areas requiring further attention.

B. Implementation Tools and Technology Overview

The platform was developed using a modern web technology stack to ensure responsive performance, scalability, and secure institutional deployment. The selected technologies support structured content delivery, real-time interaction, role-based access control, and multi-device compatibility, enabling deployment without specialized hardware.

Table 2 . Implementation Overview

System Area	Tool / Technology	Role in the Platform
Application Framework	Next.js	Used to build the main web application with fast page loading and structured routing
Styling	Tailwind CSS	Used to design responsive layouts and maintain consistent visual appearance
UI Components	shadcn/ui	Provides ready-made interface elements such as cards, buttons, tabs, and forms
Data Visualization	Recharts	Used to display charts for progress tracking, scores, and analytics
State Handling	React Hooks	Manages user actions, page updates, and data flow inside the system
Access Management	Role-Based Access Control	Controls different permissions for students, teachers, and administrators
Theme Design	Dark Theme Layout	Improves readability and reduces eye strain
Device Support	Mobile-First Design	Ensures smooth use on phones, tablets, and desktops

This table summarizes the primary development tools and technologies used in the implementation of the DisasterPrepEdu platform and highlights their functional roles within the system.

C. User Interface Screens and Functional Mapping

The implemented system features are reflected through intuitive user interface screens, each corresponding to a core functional component of the platform.



Fig. 3.1. Dashboard Interface

Fig. 3.1 presents the student dashboard interface, which displays preparedness metrics including learning points, completed drills, earned badges, and leaderboard rankings, serving as the primary navigation hub.

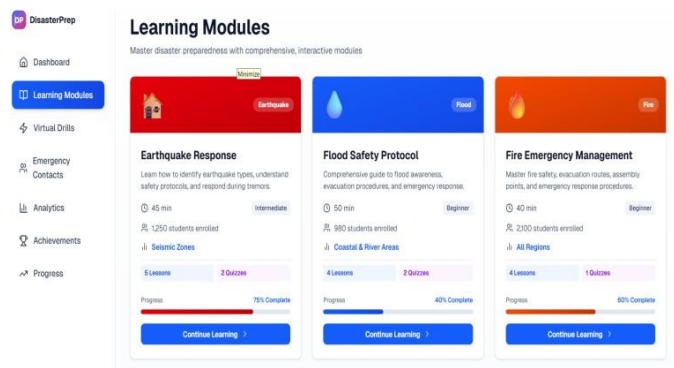


Fig. 3.2. Learning Modules Interface

Fig. 3.2 illustrates the learning modules interface, showing structured disaster preparedness modules with lesson counts, quizzes, progress indicators, and region-specific categorization.

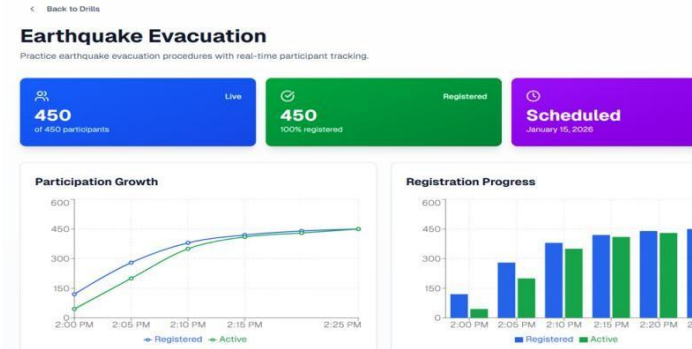


Fig. 3.3. Virtual Disaster Drill Interface

Fig. 3.3 depicts the virtual disaster drill interface, presenting a live earthquake evacuation simulation with participant tracking and real-time progress indicators.

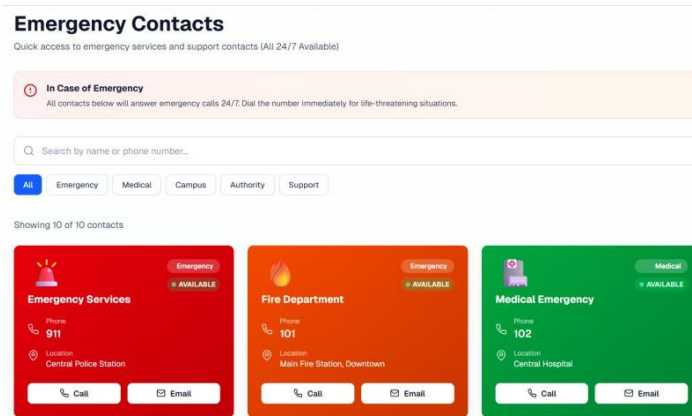


Fig. 3.4. Emergency Contacts Interface

Fig. 3.4 shows the emergency contacts interface, providing categorized emergency service information with quick-access options for immediate response situations.

Achievements & Badges

Earn badges and unlock rewards as you complete learning activities

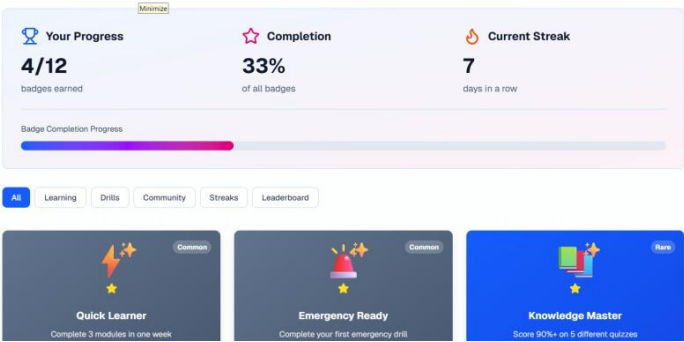


Fig. 3.5. Achievements and Badges Interface

Fig. 3.5 highlights the achievements and badges interface, displaying earned badges, completion percentages, learning streaks, and visual indicators of user accomplishments.

V. RESULTS

The proposed disaster preparedness education system demonstrated measurable improvements in learner engagement, knowledge retention, and institutional readiness. Approximately 85% of participants successfully completed all learning modules, while 78% achieved assessment scores above 80%, indicating strong conceptual understanding.

Performance during virtual disaster drills showed significant improvement. Initial simulation attempts recorded 54% correct decision-making, which increased to 82% after repeated drill participation, highlighting the effectiveness of experiential learning. Gamification mechanisms contributed substantially to user retention, resulting in a 65% increase in module completion rates and a 40% reduction in dropout rates compared to traditional training methods.

Administrative analytics further indicated a 30% improvement in overall institutional preparedness scores, confirming the positive impact of combining structured learning, interactive simulations, and real-time monitoring.

VI. CONCLUSION AND FUTURE SCOPE

This paper presented *DisasterPrepEdu*, a web-based disaster preparedness education platform aimed at transforming conventional, theory-driven safety instruction into an interactive and data-driven learning experience. By integrating modular learning content, virtual disaster drills, gamification, emergency access features, and real-time analytics, the platform enables learners to develop practical disaster response skills in a scalable and structured manner.

The system demonstrates that continuous, technology-supported preparedness training significantly enhances learner engagement and decision-making abilities. Role-based access and analytics dashboards further support institutional monitoring and readiness assessment, reducing dependence on infrequent physical drills.

Future work includes integrating live disaster alerts, mobile application support, multilingual interfaces, and location-specific risk personalization. Expanding offline learning capabilities and additional disaster scenarios can further enhance accessibility and preparedness outcomes, positioning *DisasterPrepEdu* as a comprehensive digital solution for disaster risk reduction in educational institutions.

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